WHAT SAS AND R USERS NEED TO KNOW ABOUT GLMM:

AN EXAMPLE DRIVEN DISCUSSION

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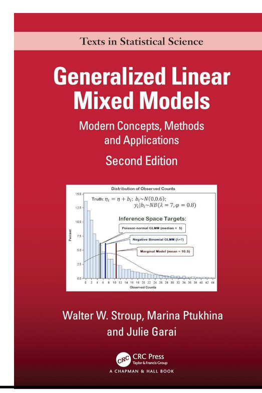
CONTEXT

- Follow up from Conference on Applied Statistics in Agriculture and Natural Resources on May 13-16, 2024 in Ames, Iowa
  - Open Discussion on the History and Future of Mixed Models
  - Workshop on troubleshooting and advanced modeling techniques with existing R packages
- May 2024 release of 2nd edition of "Generalized Linear Mixed Models: Modern Concepts, Methods and Applications", by W. Stroup, M. Ptukhina and Julie Couton
- Mixed models with a history of "silo" development… until ~1990's
- Siloing still alive and well in GLMM world?



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OBJECTIVES

1. **Open discussion** on conceptual and methodological best practices that users of generalized linear mixed models (GLMM) should know and address appropriately, regardless of choice of software platform
2. Introduce a series of 11 **selected examples** that illustrate specific GLMM issues in the context of software implementation
   - Differential software capabilities: SAS and R?
   - Open challenge
3. Propose next steps and garner interest for a **joint NCCC group effort**

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GLMM DISCUSSION:
FOCUS POINTS

- When modeling, **what do analysts need to know** regardless of software platform
  - GLMM issues are NOT intellectual exercises … rather, have very direct practical implications for estimation/inference.
- Evidence-based discussions … rather than data-free opinions
  - As per Walt Federer: *“Don’t argue… take data”*
  - Series **of selected examples** for illustration
  - Common problems in agriculture, biology, biomedical field

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GLMM DISCUSSION:
FOCUS POINTS

- As applied statisticians, we need to be able to implement best practices for each situation, regardless of software
- This is **NOT an R vs. SAS discussion**… not pushing one over the other
  - … missing the point.
- Whatever software platform is used, it needs to be **capable** of dealing with the issues and it needs to be **accessible** to users
- Selected examples worked out in SAS… some in R (mostly with limitations)… **open to contributions** from R-literate modelers

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SELECTED EXAMPLES

Sources: *SAS for Mixed Models* (Stroup, et al. 2018) and *Generalized Linear Mixed Models 2nd edition* (Stroup, et al. 2024).

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EXAMPLES

Example 1: RCBD, Gaussian response. Negative variance component estimates for block effects	Example 2: RCBD, Count response. Overdispersion. Choice of inference space and estimation method. Ripple effects on biased estimates and inferential statistics	Example 3: CRD with repeated measures, Gaussian response. Covariance structures. Bias correction.	Example 4: CRD with repeated measures, Binomial response. Same issues as Example 3 + Conditional vs. Marginal model specification. Inference space. Method of estimation.
Example 5: RCBD with substantial within-block heterogeneity due to large block size. Spatial variability and smoothing splines.	Example 6: Zero-inflated count data. Zero-inflated Poisson and Negative Binomial models. Overdispersion. Mixture distribution. Non-linear process. Inference space and estimation method.	Example 7: Continuous proportion data with meaningful zeros and/or ones. Beta hurdle model with random effects	Example 8: RCBD, Ordinal multinomial data. Cumulative logit model with random effects. Proportional odds assumption met.
Example 9: RCBD, Ordinal multinomial data. Cumulative logit model with random effects. Proportional odds assumption violated.	Example 10: GLMM-based precision analysis. Comparison between competing designs with same sample size but optimized for different objectives	Example 11: Longitudinal data with time series wave regression and standard explanatory covariates. Model with smoothing spline.	Others? GLMM-based power analyses Complex designs with multiple (3 or more) random effects

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EXAMPLES

- SAS code worked out for all
- R code worked out for some... and limited at that

Example 1: RCBD, Gaussian response. Negative variance component estimates for block effects SAS ✓ R... Mostly worked-out	Example 2: RCB, Count response. Overdispersion. Choice of inference space and estimation method. Ripple effects on biased estimates and inferential statistics SAS ✓ R... Mostly worked out / Limited	Example 3: CRD with repeated measures, Gaussian response. Covariance structures. Bias correction. SAS ✓ R... Worked out, Limited	Example 4: CRD with repeated measures, Binomial response. Same issues as Example 3 + Conditional vs. Marginal model specification. Inference space. Method of estimation. SAS ✓ R... not yet
Example 5: RCBD with substantial within-block heterogeneity due to large block size. Spatial variability and smoothing splines SAS ✓ R... not yet	Example 6: Zero-inflated count data. Zero-inflated Poisson and Negative Binomial models. Overdispersion. Mixture distribution. Non-linear process. Inference space and estimation method. SAS ✓ R... not yet	Example 7: Continuous proportion data with meaningful zeros and/or ones. Beta hurdle model with random effects SAS ✓ R... not yet	Example 8: RCBD, Ordinal multinomial data. Cumulative logit model with random effects. Proportional odds assumption met SAS ✓ R... in the works
Example 9: RCBD, Ordinal multinomial data. Cumulative logit model with random effects. Proportional odds assumption violated. SAS ✓ R... not yet	Example 10: GLMM-based precision analysis. Comparison between competing designs with same sample size but optimized for different objectives SAS ✓ R... not yet	Example 11: Longitudinal data with time series wave regression and standard explanatory covariates. Model with smoothing spline. SAS ✓ R... not yet	Others? GLMM-based power analyses Complex designs with multiple (3 or more) random effects

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NEXT STEPS:

NCCC GROUP EFFORT?

- Open challenge: contributions from the R community?
- Completing R implementation for selected examples (as much as possible)
  - Capabilities? Limitations? Recommendations?
- Working document culminating in a group publication?
  Companion textbook to Stroup et.al ?
- Articulate recommendations for curriculum design for graduate programs in statistics and data science
- Development and teaching of requisite classes/workshops for continuing ed
- Further research:
  - Small sample properties of REML-PL estimating equations vs PQL: Type I error, power, confidence interval coverage, accuracy of estimates
  - Wider range of examples

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OPEN DISCUSSION

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ADDITIONAL SLIDES

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METHODS OF ESTIMATION

Table 9. Recommended method

	Small S (few e.u./treatment)	Large S (many replications)
$f\ (y\|b)$ skewed	Quadrature is least bad—*provided the GLMM fit and the process by which the data arise are well matched*	Use quadrature—*but take care to match GLMM to data generating process to the extent possible—model misspecification matters*
$f\ (y\|b)$ at least approximately symmetric	Use RSPL	Use RSPL

Stroup and Claassen, 2020, JABES. Pseudo-Likelihood or Quadrature? What We ThoughtWe Knew, What We Think We Know, and What We Are Still Trying to Figure Out

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METHODS OF ESTIMATION

Table 5. Randomized complete block design. Poisson-gamma data generating process

Scale parameter	N blocks	Analysis model	Estimation method	Rejection rate	Coverage	mean $\hat{\lambda}$	Mean $\hat{\sigma}^2_{blk}$	Proportion converged
$\varphi = 0.25$	5	NB	RSPL	0.062	0.951	10.3	0.50	0.993
			Quad	0.128	0.917	10.4	0.41	0.991
		PN	RSPL	0.051	0.951	9.8	0.49	0.999
			Quad	0.144	0.911	9.6	0.41	0.996
	15	NB	RSPL	0.060	0.953	10.0	0.49	1.000
			Quad	0.068	0.945	10.1	0.46	1.000
		PN	RSPL	0.056	0.944	9.4	0.47	1.000
			Quad	0.077	0.913	9.1	0.46	1.000
$\varphi = 1$	5	NB	RSPL	0.136	0.914	10.3	0.49	0.953
			Quad	0.124	0.908	10.8	0.37	1.000
		PN	RSPL	0.054	0.909	7.8	0.32	0.996
			Quad	0.126	0.819	7.4	0.37	0.995
	15	NB	RSPL	0.122	0.933	9.9	0.49	0.930
			Quad	0.082	0.936	10.4	0.44	1.000
		PN	RSPL	0.050	0.800	7.1	0.42	0.999
			Quad	0.069	0.719	6.6	0.44	1.000

Rejection Rate: proportion of data sets in which H_0 : all λ_i is rejected at $\alpha = 0.05$

Coverage: proportion of 95% confidence intervals for λ_i that contain true mean ($\lambda_i = 10$)

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